Reducing property: As it can readily be oxidised, it acts as a reducing agent. It reduces chlorine into hydrochloric acid.

$$SO_2 + 2H_2O + Cl_2 \longrightarrow H_2SO_4 + 2HCl$$

It also reduces potassium permanganate and dichromate to Mn^{2+} and Cr^{3+} respectively.

$$2KMnO_4 + 5SO_2 + 2H_2O \longrightarrow K_2SO_4 + 2MnSO_4 + 2H_2SO_4$$

$$K_2Cr_2O_7 + 3SO_2 + H_2SO_4 \longrightarrow K_2SO_4 + Cr_2(SO_4)_3 + H_2O$$

Reaction with oxygen: Sulphur dioxide is oxidised to sulphur trioxide upon heating with oxygen at high temperature. This reaction is used for the manufacture of sulphuric acid by contact process.

$$SO_2(g) + O_2(g) \xrightarrow[450^0C]{V_2O_5} 2SO_3(g)$$

Bleaching action of sulphur dioxide: In presence of water, sulphur dioxide bleaches coloured wool, silk, sponges and straw into colourless due to its reducing property.

$$SO_2 + 2H_2O \longrightarrow 2\,H_2SO_4 + 2(H)$$

$$\underset{\text{Coloured}}{X} + 2(H) \longrightarrow \underset{\text{Colourless}}{XH_2}$$

However, the bleached product (colourless) is allowed to stand in air, it is reoxidised by atmospheric oxygen to its original colour. Hence bleaching action of sulphur dioxide is temporary.

Uses:

1. Sulphur dioxide is used in bleaching hair, silk, wool etc...
2. It can be used for disinfecting crops and plants in agriculture.

Structure of sulphur dioxide:

In sulphur dioxide, sulphur atom undergoes sp^2 hybridisation. A double bond arises between S and O is due to pπ- dπ overlapping.

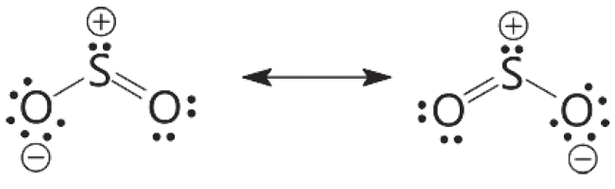


Figure 3.9 Structure of sulphur dioxide.

3.2.2 Sulphuric acid: (H_2SO_4)

Preparation:

Sulphuric acid can be manufactured by lead chamber process, cascade process or contact process. Here we discuss the contact process.

Manufacture of sulphuric acid by contact process:

The contact process involves the following steps.

i. Initially sulphur dioxide is produced by burning sulphur or iron pyrites in oxygen/air.

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